

data is available. This can be done by using the 'Serial.available' function that checks how many bytes are available at the serial port. It's generally used along with if condition check.

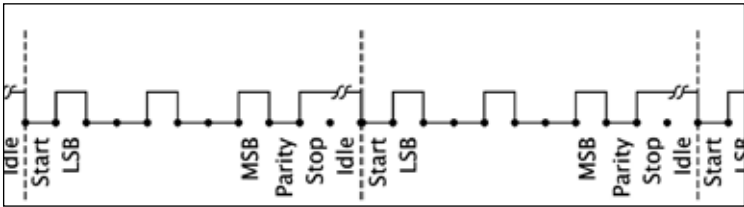
- `if ( Serial.available()) { }`

If no data is available, this will return 0 and the program won't enter the 'if' part of the program.

Serial.read

This command reads the incoming serial data one byte at a time. It's used along with serial.available to ensure that the data has been received before it's read and stored. The syntax of the function is:

- `storage_var = Serial.read ();`



Sequence for serial data transmission

IMAGE CREDIT: KATZENMEUK

Code:

```
• int incomingByte = 0;    // variable for storing incoming
serial data byte
• void setup() {
    Serial.begin(9600);    // opens serial port, set-
ting baud rate to 9600 bps
}
void loop() {
    if (Serial.available() > 0) {    // checks if data
has been received
        int incomingByte = Serial.read(); // reads
the incoming byte
    }
}
```

What if my sketch doesn't work correctly?

If your code doesn't work as expected, the only way to make it work is to